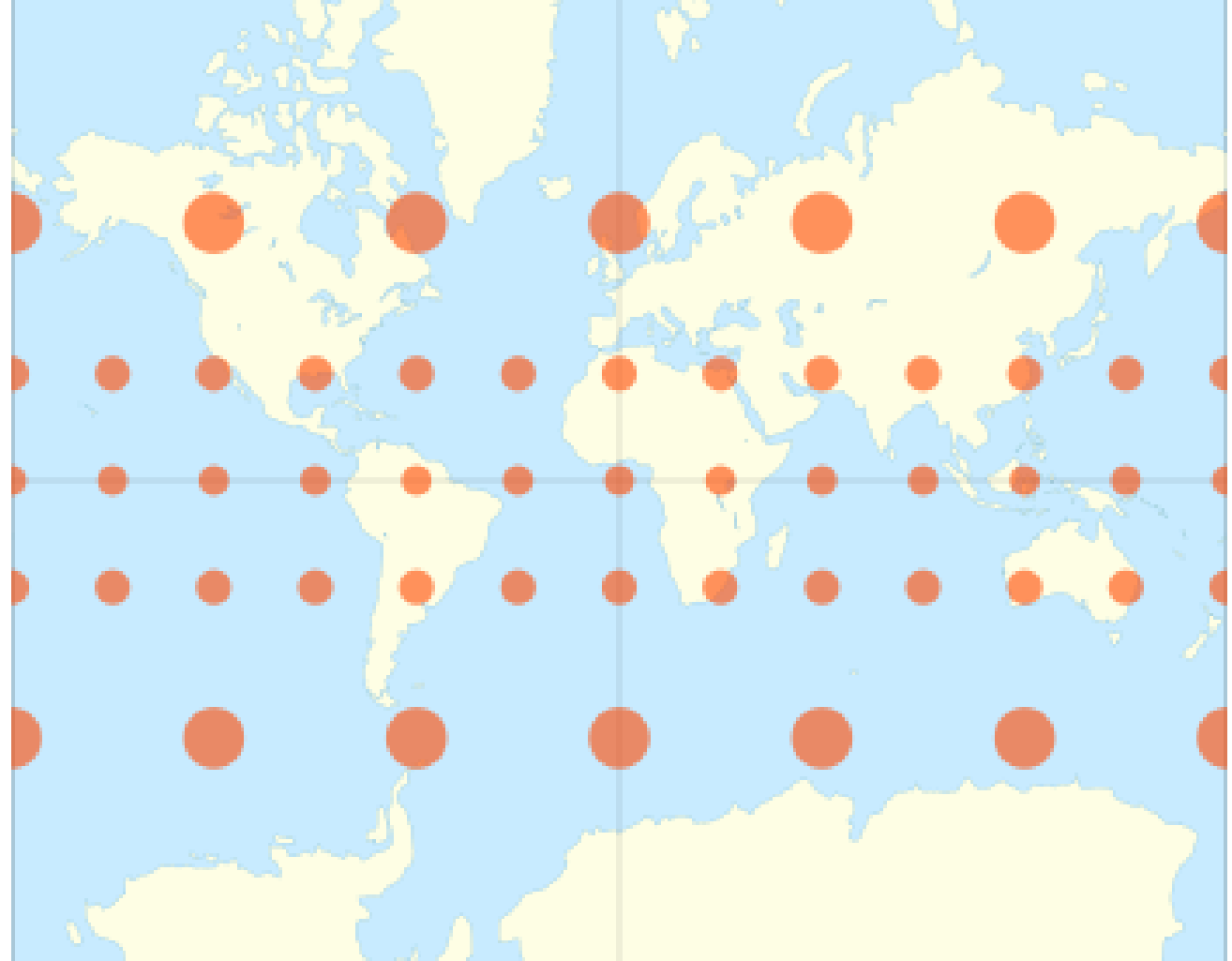
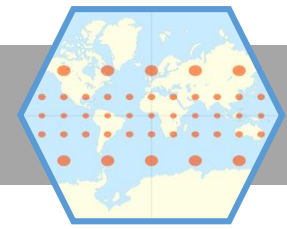


LAB 4: Cartography





EX 1: modulus of linear deformation

Plotting the modulus of linear deformation, considering these cases:

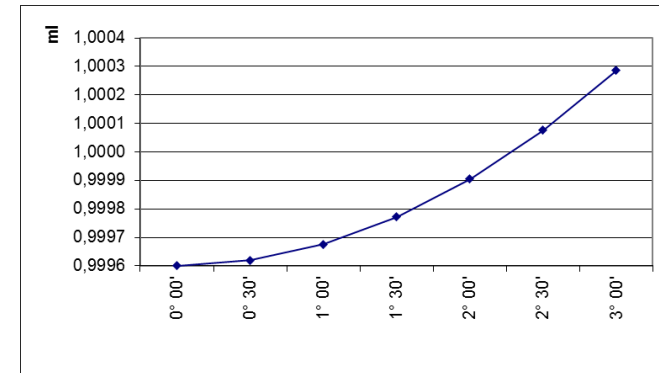
$$m_l = 0,9996 \left(1 + \frac{\lambda^2}{2} \cos^2 \varphi \right)$$

Case 1: lat= 30° long': 0°- 3°, with 0,5° of resolution

Case 2: lat= 37° long': 0°- 3°, with 0,5° of resolution

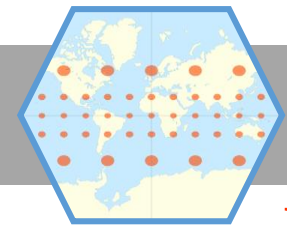
Case 3: lat= 45° long': 0°- 3°, with 0,5° of resolution

Case 4: lat= 60° long': 0°- 3°, with 0,5° of resolution



To compare the results.





EX 2: HIRVONEN equation

Using the Hirvonen equation, to transform these coordinates from geographic to cartographic.

P1 → lat = 45° 3' 45,717'' long = 7° 47' 26,292''

SR = WGS84 - UTM

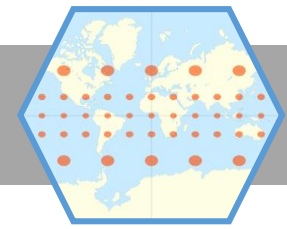
Central meridian = 9° UTM-ZONE= 32

P2 → lat = 38° 32' 34,649'' long = 16° 50' 06,493''

SR = WGS84 - UTM

Central meridian = 15° UTM-ZONE= 33





EX 2: HIRVONEN equation

$$x = R_p \operatorname{arcsenh} \frac{\cos \xi \tan \lambda'}{v}$$

$$y = a (A_1 \xi - A_2 \sin 2\xi + A_4 \sin 4\xi - A_6 \sin 6\xi)$$

where

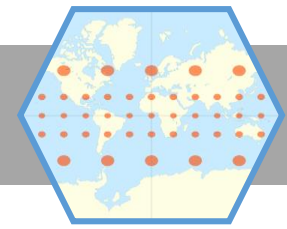
$$R_p = \frac{a^2}{c} \quad \text{Radius of polar curvature}$$

$$\lambda' = \lambda - \lambda_{mc} \quad \xi = \operatorname{arctg} \frac{\tan \varphi}{\cos(v_1 \lambda')}$$

$$v_1 = \sqrt{(1 + e'^2 \cos^2 \varphi)} \quad v = \sqrt{(1 + e'^2 \cos^2 \xi)}$$

$$\lambda' = (\lambda_P - \lambda_{mc}) \quad \text{Where } \lambda_{mc} \text{ is the central meridian of the fuse}$$





EX 2: HIRVONEN equation

$$A_1 = 1 - \frac{e^2}{4} - \frac{3e^4}{64} - \frac{5e^6}{256}$$

$$A_2 = \frac{3e^2}{8} + \frac{3e^4}{32} + \frac{45e^6}{1024}$$

$$A_4 = \frac{15e^4}{256} + \frac{45e^6}{1024}$$

$$A_6 = \frac{35e^6}{3072}$$

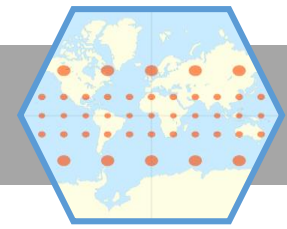
$$\text{East} = x * m_c + \text{False East}$$

$$\text{North} = y * m_c$$

where m_c = contraction modulus 0,9996

*False East = depend for RS





EX 3: HIRVONEN equation (inverse)

Using the Hirvonen equation, to transform these coordinates from cartographic to geographic.

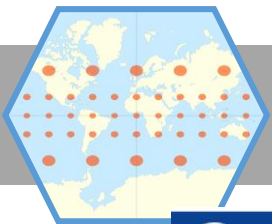
P → **East** = 470139,66 m **North** = 5031468,37 m

SR = WGS84 – UTM-ZONE 32

Long = 8° 37' 5",6209

Lat = 45° 26' 9",9617





EX 4: Copernicus images

Copernicus BROWSER EN ▼ Marco Piras ▼

VISUALISE SEARCH

SH DASHBOARD WORKSPACE

DATE: SINGLE

< YYYY-MM-DD > 30%

Show latest date ↗

Find products for current view

CONFIGURATIONS: Default ▼

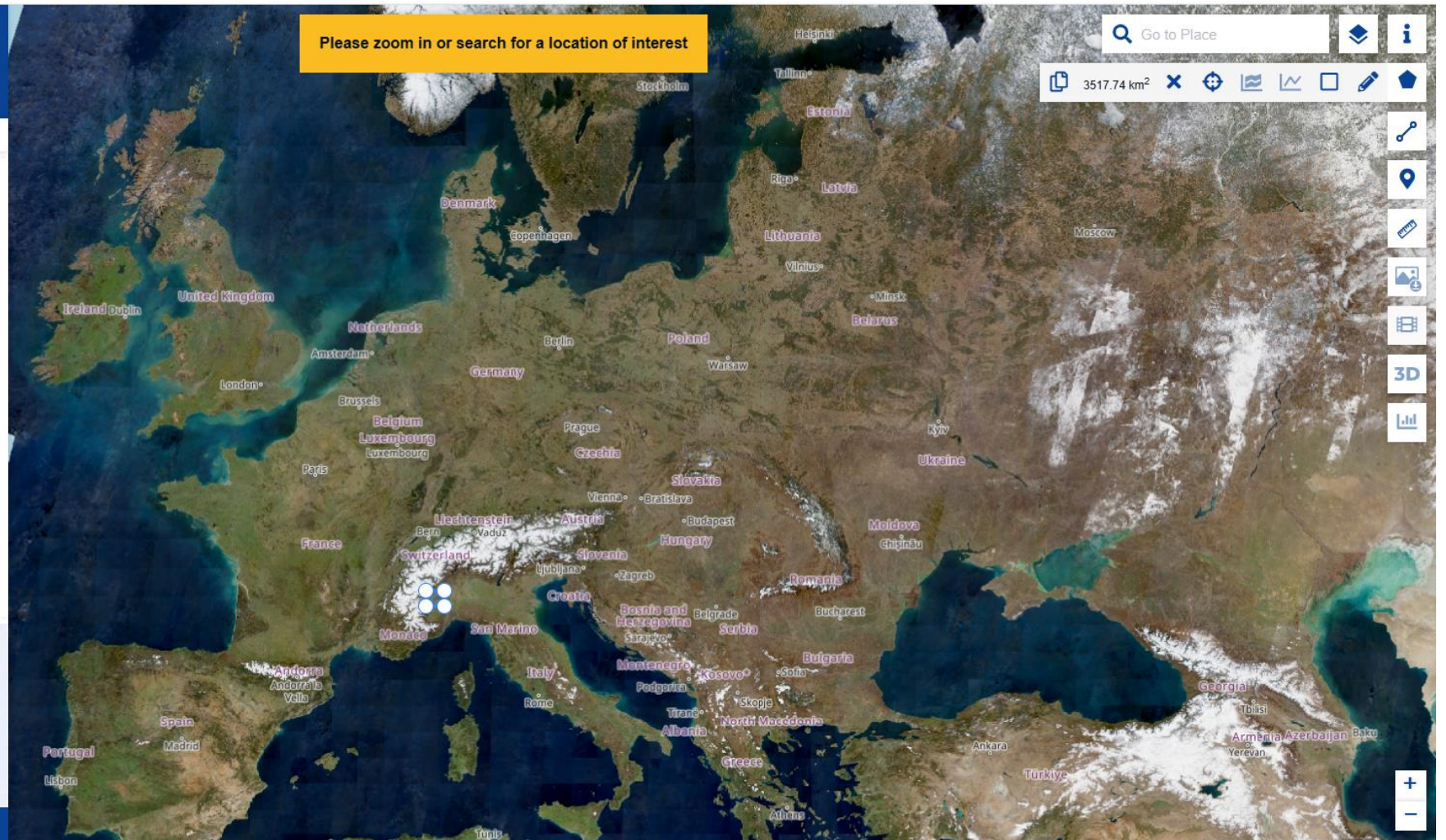
DATA COLLECTIONS:

Sentinel-2 ▼

Sentinel-2 L1C

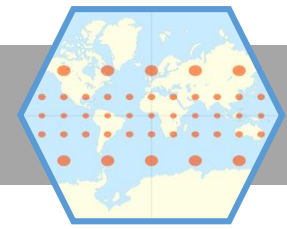
Sentinel-2 L2A ✓

European Union Copernicus ESA About Support



<https://browser.dataspace.copernicus.eu/>





EX 4: Copernicus images

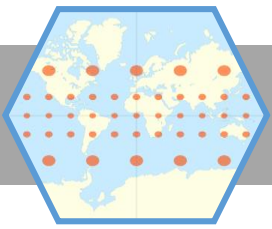
- **To Create a AOI and export the following products in two different year (2025 and 2020)**
 - **NDVI**
 - **Air quality (CO2 or Methane)**
 - **True color**
 - **False Color**

<https://browser.dataspace.copernicus.eu/>





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EX 4: Copernicus images

Basic **Analytical** High-res print Download ×

Image download

Image format:

TIFF (32-bit float)

Image resolution:

HIGH

2500 x 477 px

Coordinate system:

WGS 84 (EPSG:4326)

Resolution:
lat.: 0.0011065 deg/px (4.0sec/px)
long.: 0.0015656 deg/px (5.6sec/px)

Clip extra bands

☒

Layers:

Visualised

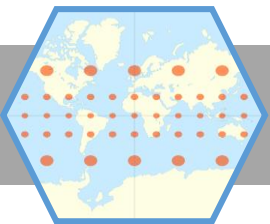
☒ True color
☐ False color
☐ NDVI

Raw

☐ B01
☐ B02
☐ B03
☐ B04
☐ B05
☐ B06
☐ B07
☐ B08
☐ B8A
☐ B09

Show more

Preview



EX 4: Copernicus images

- To create a TIMELASPE considering at least 6 years, about 2 different themes.

Timelapse

2025-04-03 - 2025-05-14

Filter by months

Select 1 image per:

☐ orbit ☒ day ☐ week ☐ month ☐ year

Sentinel-2 L2A: True color

[Search](#)

[Add layers from pins](#)

Visualisations

Min. tile coverage: 0%

Max. cloud coverage: 39%

☒ Select All

5 km

2025-04-23

Copernicus
Data Space Ecosystem

Speed: 1 fps Transition: None

6 / 8

[Download](#)

